

# Sera Nolin

## Rendering Researcher

Research-oriented graphics programmer investigating visibility, material representation, GPU scheduling, and practical renderer architecture.

**loc:** Lakeview Research Campus

**env:** Linux

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**git:** github.com/fake-sera-nolin

### /// SKILLS

#### Build:

CMake Ninja Git Linux

#### Graphics:

Vulkan OpenGL GLSL SPIR-V PBR  
Shadow Mapping

#### Mathematics:

Linear Algebra Geometry Probability

#### Programming:

C++ C Python Lua

#### Research:

Benchmark Design Experiment Tracking  
Technical Writing

### /// EDUCATION

#### MSc Computer Graphics

Fictional Academy of Visual Computing

**Status:** Research completed

#### BSc Software Systems

Imaginary Polytechnic

**Status:** First Class

#### Advanced Diploma in Programming

Northstar Technical College

**Status:** Completed

### /// RESEARCH

#### Focus:

GPU resource scheduling  
Material systems Visibility  
Frame graphs

#### Methods:

Controlled benchmarks GPU captures  
Micro-tests

#### Output:

Technical reports Reproducible scenes  
Performance tables

### /// TALKS

**Event:** Fictional Graphics Meetup

**Topic:** Thinking in GPU Resource Lifetimes

**Event:** Imaginary Rendering Workshop

**Topic:** From Pass Declarations to a Working Frame

### /// PROJECTS

#### Visibility Atlas (Research Engineer)

Synthetic benchmark investigating visibility-buffer layouts across scenes with highly different triangle and material distributions.

C++ Vulkan Compute Benchmarking

#### Lumenless (Rendering Researcher)

Fictional renderer prototype comparing tiled lighting and clustered-light approaches under controlled workloads.

C++ GLSL Vulkan Lighting

#### Shader Observatory (Tooling Engineer)

Offline shader inspection tool that groups reflection metadata and reports resource usage per pipeline.

C++ SPIR-V Reflection

#### Mesh Census (Pipeline Developer)

Asset-analysis utility for measuring topology, UV density, material count, and texture footprint in synthetic datasets.

C++ Assimp CLI